

Math Primer

(3. Complex Numbers)

Contents



- ▶ Polynomial Roots
- ▶ Introduction to Complex Number
- ▶ Algebra of Complex Numbers
- ▶ Polar Representation
- ▶ Important Results



Euler
(1707-1783)

Polynomials and Roots

- ▶ A 'n' degree polynomial is represented as

$$f(x) = \underbrace{a_n}_{+} x^{\textcircled{n}} + \underbrace{a_{n-1}}_{+} x^{n-1} + \dots + \underbrace{(a_0)}_{+} x^{\textcircled{0}}$$

- ▶ The highest power of x is the degree of the polynomial
- ▶ e.g., $f(x) = \underbrace{ax^{\textcircled{1}}}_{+} + \underbrace{b}_{+}$ (degree 1 or 'linear') , $f(x) = \underbrace{ax^{\textcircled{2}}}_{+} + \underbrace{bx}_{+} + \underbrace{c}_{+}$ (degree 2 or 'quadratic')
 - ▶ The number of solutions to $f(x) = 0$ are called the roots of the polynomial
 - ▶ An 'n' degree polynomial will have 'n' roots , e.g.

- ▶ A linear polynomial has 1 root $x = \underline{-b/a}$,
- ▶ A quadratic polynomial has 2 roots $x = \underline{\underline{\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}}$

$$\underline{ax+b=0}, \quad x = \underline{-b/a}$$

Special Case of Quadratic Function

► A quadratic function has 2 roots

- If $D = (b^2 - 4ac) > 0$, then it has two 'real roots' (function crosses X axis at 2 distinct points)
- If $D = (b^2 - 4ac) = 0$, then it has one repeated real root (function touches X axis at a point)
- If $D = (b^2 - 4ac) < 0$, then it has no real roots (function never crosses X axis)

$$x^2 - 4x + 3$$

$$(x-1)(x-3) = 0$$

$$\alpha, \beta = \frac{-b \pm \sqrt{D}}{2a}$$

$$(x-1)^2$$

$$2^2 - 4 \cdot 1 \cdot (-8) = 36$$

$$(x-2)^2 = 0$$

$$2^2 - 4 \cdot 1 \cdot 4$$

$$D \leq 0$$



$$f(x) = x^2 + 2x - 8$$

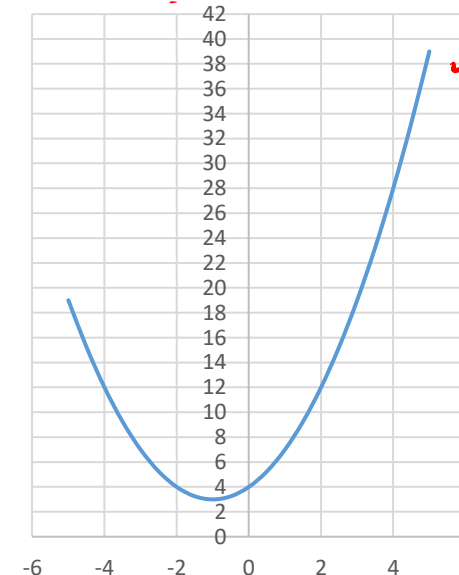
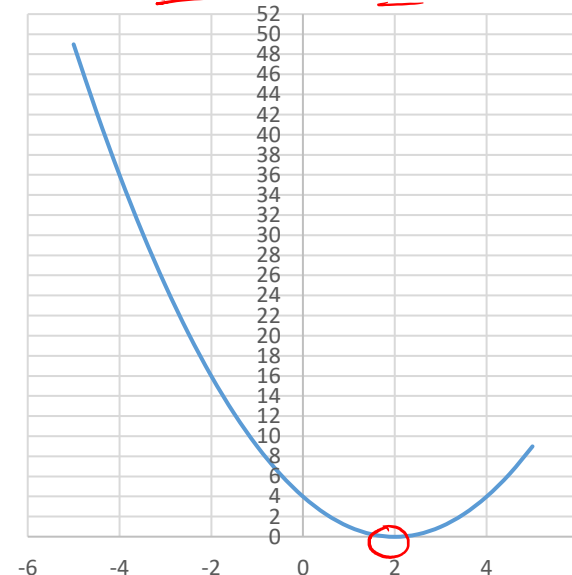
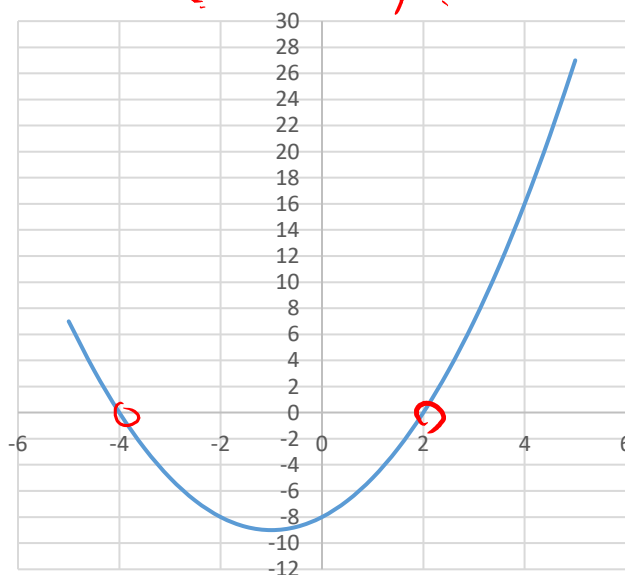
$$f(x) = x^2 - 4x + 4$$

$$f(x) = x^2 + 2x + 4$$

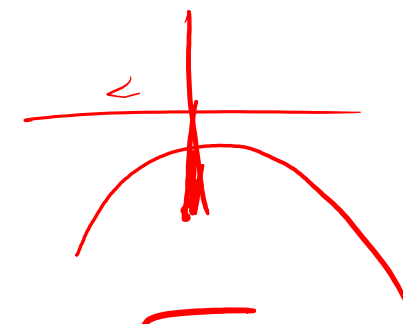
$$D = 36, \text{ roots} = (2, -4)$$

$$D = 0, \text{ roots} = (2, 2)$$

$$D = -12, \text{ roots} = ? ?$$



$$\sqrt{-1} \rightarrow$$

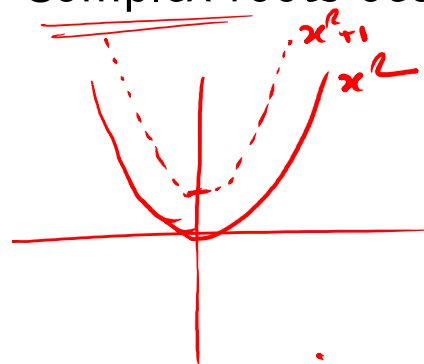


$$\sqrt{-8}$$

$$\sqrt{8 \cdot (-1)} = \sqrt{8} \cdot \sqrt{-1}$$

Complex Numbers

- ▶ Let's call $i = \sqrt{-1}$ as the imaginary number (Some books adopt the notation j)
- ▶ Note the following properties : $i^2 = -1$, $i^3 = -i$, $i^4 = 1$ $i^3 = i^2 \cdot i = (-1) \cdot i = -i$
- ▶ A complex number is written as $(a + ib)$, where a represents the real part and b the imaginary part
- ▶ e.g., $(2 + 3i)$, $\text{Re}(2+3i) = 2$, $\text{Im}(2+3i) = 3$
- ▶ Then polynomials not having real roots can be seen as having imaginary roots
- ▶ e.g., $x^2 + 1 = 0$ has roots $(i, -i)$ and $x^2 + 2x + 2$ has roots $(-1 + i, -1 - i)$
- ▶ If $z = a + ib$, then conjugate of z is denoted as $\bar{z} = a - ib$
- ▶ Complex roots occur in conjugate pairs



$$i = 0 + 1 \cdot i$$

$$-i = 0 - 1 \cdot i$$

$$x^2 + 1 = 0$$

$$x^2 = -1$$

$$\Rightarrow x = \pm \sqrt{-1} \quad \underline{\underline{\pm i}}$$

$$(a + ib), (a - ib)$$

$$\underline{\underline{(-i)^2}} = (-i) \cdot (-i)$$

$$= i \cdot i = i^2 = -1$$

$$(2i)^2 = 4 \cdot i^2 = -4$$

$$(x+1)^2 + 1$$

$$x^2 + 2x + 2 = 0$$

$$\frac{-2 \pm \sqrt{4 - 8}}{2}$$

$$= \frac{-2 \pm 2i}{2}$$

$$= (-1 \pm i)$$

Algebra of Complex Numbers

► Addition / Subtraction :

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 6 \\ 3 \end{bmatrix} = \begin{bmatrix} 8 \\ 6 \end{bmatrix} \rightarrow$$

$$a+ib$$

$$z_1 = \underline{a} + ib, z_2 = \underline{c} + id, \text{ then } \underline{z_1 + z_2} = \underline{(a + c) + i(b + d)}, \underline{z_1 - z_2} = \underline{(a - c) + i(b - d)}$$

$$\text{e.g., } z_1 = \underline{6 + 3i}, z_2 = \underline{2 - i}, z_1 + z_2 = \underline{8 + 2i}, z_1 - z_2 = \underline{4 + 4i}$$

► Multiplication :

$$1 = 1 + 0i$$

$$z_1 = a + ib, z_2 = c + id, \text{ then } z_1 z_2 = (ac - bd) + i(ad + bc)$$

$$\text{e.g., } z_1 = \underline{3 + 5i}, z_2 = \underline{2 + 6i}, z_1 z_2 = \underline{-24 + 28i}$$

► Inverse :

$$z = \underline{a + ib}, z^{-1} = \frac{1}{z} = \frac{a}{a^2 + b^2} + i \frac{-b}{a^2 + b^2} \leftarrow$$

$$\text{e.g., } z = \underline{2 - i}, z^{-1} = \frac{\underline{2+i}}{\underline{5}}$$

$$z^{-1} = \frac{\bar{z}}{|z|^2}$$

$$(a+ib)(c+id)$$

$$= ac + iad + ibc - bd$$

$$= (ac - bd) + i(ad + bc)$$

► Division :

$$z_1 = \underline{6 + 3i}, z_2 = \underline{2 - i}, \text{ then } \frac{z_1}{z_2} = (6 + 3i) \left(\frac{1}{2 - i} \right) = (6 + 3i) \left(\frac{2+i}{5} \right) = \frac{(9+12i)}{5}$$

$$\sqrt{-1}$$

$$i$$

$$\sqrt{i}$$

Complex Plane

► **Modulus :** mod z $\sqrt{a^2 + b^2}$ $\in \mathbb{R}$

$z = a + ib$, $|z| = \sqrt{a^2 + b^2}$, called the modulus which is the length of the vector

Conjugate $z = \underline{a} - ib$ is the mirror image on z on X-axis

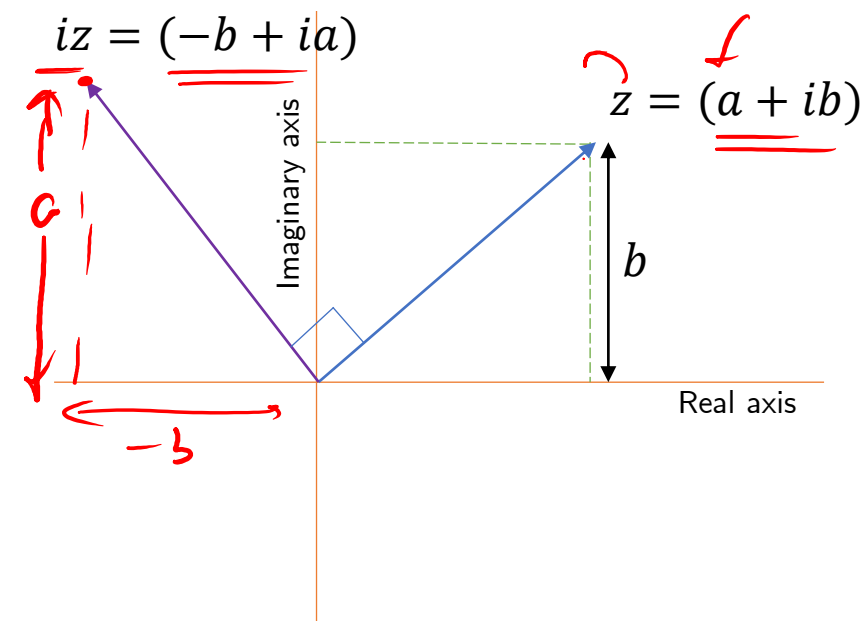
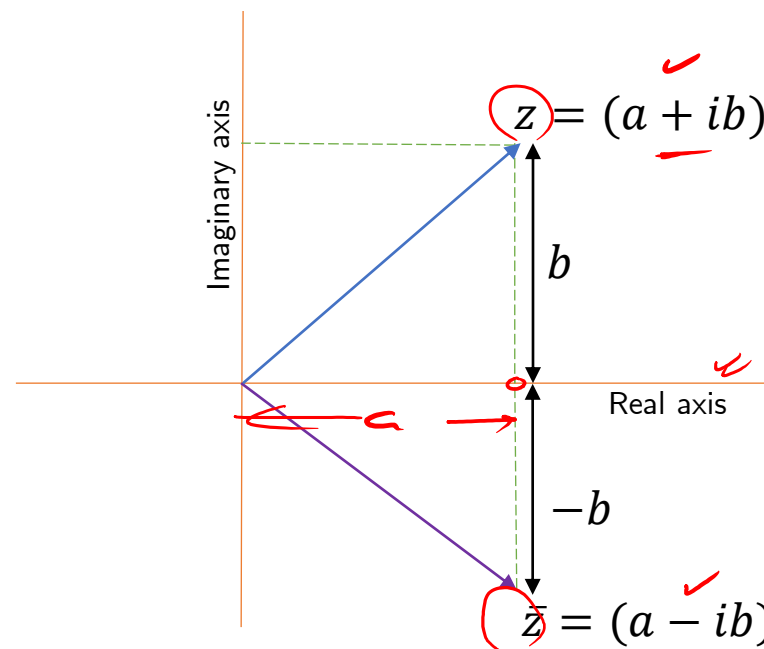
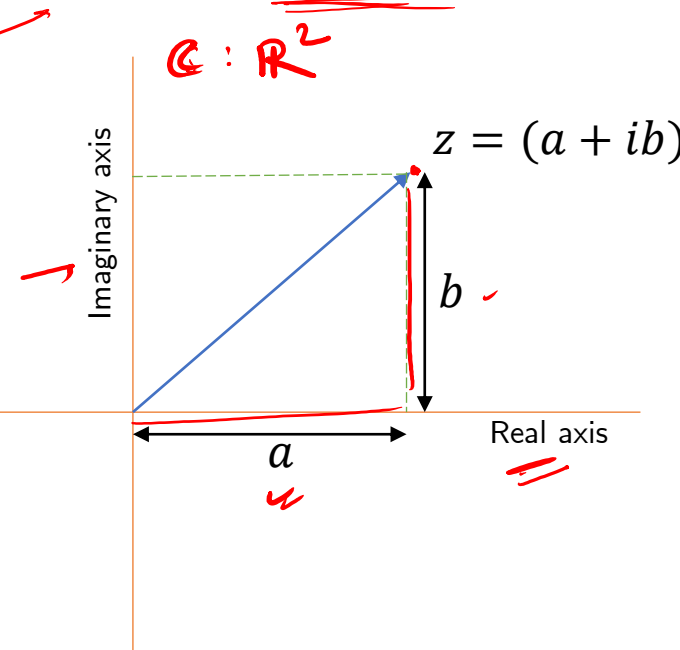
Also, $z\bar{z} = |z|^2$

Multiplying i with z , rotates it by 90° anti-clockwise

$$\sqrt{a^2 + (-b)^2} = \sqrt{a^2 + b^2}$$

$$\begin{aligned} (a+ib)(a-ib) &= a^2 - (ib)^2 \\ &= a^2 + b^2 = |z|^2 \end{aligned}$$

$$\begin{aligned} ia + i^2b &= (-b + ia) \end{aligned}$$

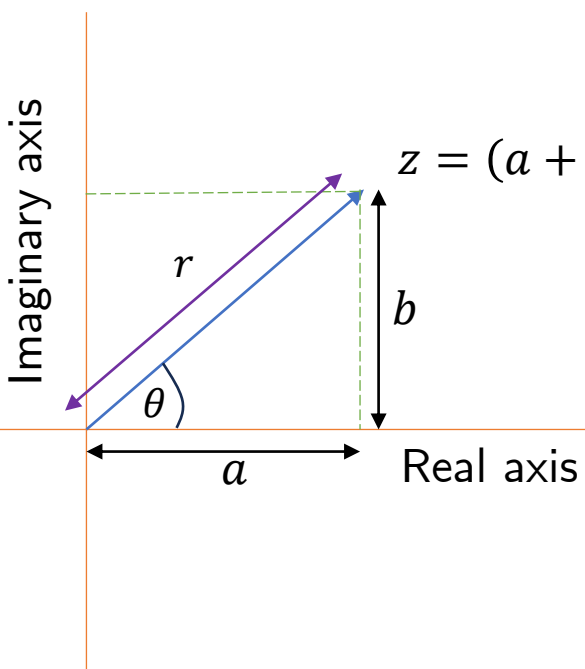


Polar Representation

- A complex number can be represented with the length and the angle it makes with X axis :

$z = \underline{a} + \underline{ib} = r(\cos \theta + i \sin \theta) = re^{i\theta}$, where $r = |z| = \sqrt{a^2 + b^2}$, θ is the angle with x-axis which is called as the argument of z given as $\theta = \tan^{-1} \left(\frac{b}{a} \right)$

$e^{i\theta} = \cos \theta + i \sin \theta$ is called as Euler Identity



$$z = (a + ib) = \underline{r(\cos \theta + i \sin \theta)} = \underline{re^{i\theta}}$$

$$\underline{(a+ib)} \leftrightarrow \underline{re^{i\theta}}$$

$$\underline{r = |z|} \\ \underline{r = \sqrt{a^2 + b^2}}, \quad \underline{\theta = \tan^{-1}(b/a)}$$

Important Results

Euler's Identity

De Moivre's Theorem

Power Series

$$\begin{aligned}
 & \left. \begin{aligned}
 & e^{i\theta} = \cos \theta + i \sin \theta \\
 & e^{-i\theta} = \cos \theta - i \sin \theta \\
 & e^{i\pi} + 1 = 0 \\
 & (\cos \theta + i \sin \theta)^n = e^{in\theta} = \cos(n\theta) + i \sin(n\theta)
 \end{aligned} \right\} \begin{aligned}
 & \cos(-\theta) = \cos(\theta) \\
 & \sin(-\theta) = -\sin(\theta) \\
 & \cos(\pi) + i \sin(\pi) = -1 + i \cdot 0 = -1
 \end{aligned} \\
 \\
 & \left. \begin{aligned}
 & (e^{i\theta})^n = e^{i(n\theta)} = \cos(n\theta) + i \sin(n\theta) \\
 & e^{\pm z} = 1 \pm z + \frac{1}{2!} z^2 \pm \frac{1}{3!} z^3 + \dots \\
 & \sin z = z - \frac{1}{3!} z^3 + \frac{1}{5!} z^5 - \dots \\
 & \cos z = 1 - \frac{1}{2!} z^2 + \frac{1}{4!} z^4 - \dots
 \end{aligned} \right\} \begin{aligned}
 & e^z \\
 & e^{-z}
 \end{aligned}
 \end{aligned}$$

Exercises



1. If $\underline{(x + iy)^3} = \underline{u + iv}$, then show that $\frac{u}{x} + \frac{v}{y} = 4\underline{(x^2 - y^2)}$
2. If $\underline{\left(\frac{1+i}{1-i}\right)^m} = 1$, then find the least m
3. Find $z \in \mathbb{C}$ such that $z^2 \in \mathbb{R}$
4. Prove that there is no complex number z , such that $|z| - z = i$
5. Express $\ln(a + ib)$ in standard form